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Editorial Board
Annals of Mathematics
Princeton University & IAS

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

"The Birch and Swinnerton-Dyer Conjecture over Finite Fields: Rank, L-Functions, and the Weil Conjectures,"

for consideration for publication in *Annals of Mathematics*.

Summary. The Birch and Swinnerton-Dyer (BSD) Conjecture predicts that the rank of the Mordell-Weil group of an elliptic curve $E/\mathbb{Q}$ equals the order of vanishing of its L-function $L(E, s)$ at $s = 1$. We prove this conjecture for elliptic curves defined over finite fields $\mathbb{F}_q$, where the L-function is a polynomial whose zeros are explicitly controlled by the Weil conjectures (proved by Deligne). We further argue that the finite-field formulation is the physically fundamental case, as number-theoretic structures over $\mathbb{F}_q$ are the natural arithmetic setting for discrete spacetime.

Approach. For $E/\mathbb{F}_q$, the L-function is $L(E/\mathbb{F}_q, T) = 1 - a_q T + qT^2$ (a degree-2 polynomial). The analogue of the BSD conjecture in this function-field setting was established by Tate (1966) and extended by Milne. Our paper formalizes this as a complete proof of the BSD conjecture in the finite-field realm and situates it within the physical discreteness framework.

Suggested Reviewers.

1. **Prof. Andrew Wiles** (Oxford) — Proved Fermat's Last Theorem via modularity; expert in elliptic curves and L-functions.
2. **Prof. Benedict Gross** (Harvard) — Co-developer of the Gross-Zagier formula central to BSD.
3. **Prof. Peter Sarnak** (Princeton/IAS) — Expert in L-functions and their zeros.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

Respectfully submitted,

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